Open problems:
Quantum pure states are not enough

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first. Items marked [OPTIONAL] would strengthen a result that is already checked; no claim of the brief waits on them.

Theorem 5 is verified, and Theorem 3 for $\dim \mathcal{H} \geq 2$. Example 2 is verified at the level of topology and measure and, on the Riemann sphere, of the symplectic form and of Hamiltonian evolution, with the correction of the item "Hamiltonian evolution on the classical qubit". Theorem 1 is not: see the two items marked below.

From the brief

1. [BLOCKS VERIFICATION: BRIEF ISSUE] **The range of Theorem 1.** The theorem claims consistency for $1 \leq \dim V < \infty$, while its proof uses $2 \leq \dim X$. Making it precise needs four choices.

   - *Dimension* 0. The range starts at 1 if a 0-dimensional symplectic manifold ($n = 0$ degrees of freedom) is allowed, and at 2 if not. For $\dim V = 1$ the state space is a single point, and $P(V)$ has two points exactly when $2 \leq \dim V$ (`subsingleton_projectivization`, `nontrivial_projectivization_iff`).
   - *Theorem 3 depends on it.* If a point is allowed, Theorem 3 is false for $\dim \mathcal{H} = 1$: the CST ensembles of a point and the density operators of $\mathbb{C}$ are both a single point, and the constant map identifies them (`cst_qeDens_consistent_of_finrank_one`). So the brief cannot keep both $1 \leq \dim V$ in Theorem 1 and Theorem 3 for every $\mathcal{H}$.
   - *The empty space.* For $\dim V = 0$, $P(V)$ is empty (`isEmpty_projectivization`), which is a symplectic manifold of every dimension. Either lower bound, 1 or 2, needs CST to exclude it, for instance by requiring an ensemble.
   - *What "represented by" keeps.* If X and $P(V)$ need only be in bijection, the "only if" is false: $P(\ell^2)$ and the sphere $P(\mathbb{C}^2)$ both have $\mathfrak{c}$ points (`nonempty_projectivization_equiv`), and the sphere's symplectic structure can be carried over. The theorem needs V Hausdorff and X to carry the quotient topology of $P(V)$; finiteness then comes from CST's Liouville measure ω^n with finite n, not from X being symplectic (Log, "Problems found").

   Proposed: exclude dimension 0 and the empty space, so that Theorem 1 reads $2 \leq \dim V < \infty$ for a Hausdorff V with $X = P(V)$ as topological spaces. Theorem 3 and its Lean proof then stand as they are. The alternative keeps $1 \leq \dim V$, also excludes the empty space and restricts Theorem 3 to $\dim \mathcal{H} \geq 2$. (Log, "Problems found", first three items.)

2. **Ensembles supported on a subspace.** The proof of Theorem 3 compares the ensembles with support in a two-dimensional subspace A. For $\dim \mathcal{H} \geq 3$ no CST ensemble has support in $P(A)$, which has Liouville measure zero. Whether the brief means the classical ensembles of the subsystem $P(A)$, with its own Liouville measure, is open. The formalization compares the whole ensemble spaces instead. (Log, "Problems found", second item.)

3. **Hamiltonian evolution on the classical qubit.** Example 2 identifies Hamiltonian evolution with the diffeomorphisms that conserve the volume. Area-preserving reflections of the sphere are not Hamiltonian, so the area form should be meant. Lean checks this for complex conjugation of the Riemann sphere, the Bloch reflection $a_1 \mapsto -a_1$ (`conjSphere_preserves_area_reverses_form`, `globalFlow_ne_conjSphere`, `timeDependentFlowOf_ne_conjSphere`). With the area form the claim is checked for evolutions. A vector field preserves the Fubini–Study form exactly when it is Hamiltonian (`lieDeriv_areaForm_eq_zero_iff_hamiltonian`), and a jointly smooth time-dependent flow preserves it exactly when every velocity field is Hamiltonian (`preserves_areaForm_iff_hamiltonian`), as does a smooth isotopy from the identity through its generating field (`isotopy_preserves_areaForm_iff_hamiltonian`). No flux homomorphism is needed: the obstruction is a class in $H^1_{dR}(S^2)$, which vanishes because the sphere is simply connected. Open is the statement for single maps, that every orientation-preserving, area-preserving diffeomorphism of the sphere is the end of a Hamiltonian evolution. It needs Smale's theorem $\mathrm{Diff}^+(S^2) \simeq SO(3)$, or at least that $\mathrm{Diff}^+(S^2)$ is connected through smooth isotopies, and a Moser argument for area forms on S^2. That quantum evolutions are exactly the rotations is checked (`exists_unitaryAction_eq_iff`). (Log, "Problems found", Example 2 item.)

From the formalization

1. [BLOCKS VERIFICATION: FORMALIZATION GAP] **Theorem 1.** Neither direction of Theorem 1 is formalized: that $P(V)$ with the Fubini–Study form is symplectic for $\dim V < \infty$, and that a symplectic $P(V)$ forces $\dim V < \infty$. TauCeti has symplectic manifolds, the topology of projective space and the Fubini–Study form on the Riemann sphere $\cong P(\mathbb{C}^2)$; the Fubini–Study form on $\mathbb{C}P^n$ for $n \geq 2$ and the Liouville measure of a symplectic form are still missing upstream. Theorem 3 for $\dim \mathcal{H} = \infty$ rests on the second direction, so it is formalized only for $\dim \mathcal{H} < \infty$. The second direction needs the topology of $P(V)$; as sets it is false (`nonempty_projectivization_equiv`). (Log, "Results".)

2. **Example 2.** $P(\mathbb{C}^2)$ is homeomorphic to the sphere (`nonempty_homeomorph_sphere`), its area is $dz\, d\theta$ (`toSphere_eq_dz_dtheta`), the quantum evolutions are the rotations (`exists_unitaryAction_eq_iff`) and every Hamiltonian makes the Bloch vector precess (`exists_blochVector_precession`). Still missing: the area as the Liouville measure of the Fubini–Study form, and $\omega = dz \wedge d\theta$ in the chart, which need a Liouville measure upstream; the diffeomorphism $\mathbb{C}P^1 \cong S^2$ that compares the Fubini–Study form with the round area form; the integrated rotation of the precession; the classical side of linear Hamiltonians, $X_{n \cdot x} = n \times x$, which needs moment maps. Refinements of Hamiltonian evolution are in the next item. (Log, "Differences".)

3. [OPTIONAL] **Refinements of Hamiltonian evolution.** The evolutions of Example 2 are checked with each velocity field X_t Hamiltonian for its own H_t (`preserves_areaForm_iff_hamiltonian`, `isotopy_preserves_areaForm_iff_hamiltonian`). Three refinements remain.

- *One jointly smooth $H(t, x)$.* TauCeti builds the primitive of a closed 1-form by lifting through a covering space, separately for each t, and has no version with parameters. A route: normalize $H_t(x_0) = 0$; prove a Poincaré lemma with a time parameter in charts, where the flat step is the homotopy operator `poincareHomotopy` made jointly smooth with `contDiffOn_integral_smul_of_contDiffOn`; then the points y at which $t \mapsto H_t(y)$ is smooth form an open and closed set, and H is jointly smooth near each point. The chart step repeats TauCeti's manifold Poincaré lemma with a parameter and needs chart expressions of $\iota_{X_t}\omega$; estimated at 400–700 lines.
- *Area instead of area form along an evolution.* Along an isotopy from the identity, $\varphi_t^*\omega = \pm\omega$ forces the sign $+$, since it cannot jump; with area preservation stated pointwise this is a short continuity argument. The measure-theoretic statement needs the Liouville measure of ω, which is missing upstream (item 2).
- *Conservation of entropy.* The brief's "and therefore the entropy is conserved" follows from measure preservation, since the Gibbs entropy is invariant under measure-preserving bijections. For the rotations this is direct; for a general Hamiltonian flow it needs the Liouville measure, as above.

4. **Extreme points.** The proof of Theorem 5 starts from the fact that the pure states are the extreme points of $D(\mathcal{H})$. The formalization does not need it and does not prove it. (Log, "Differences", last item.)